FLAT PULLBACK OF RELATIVE DIMENSION E IS THE BIVARIANT ACTION OF THE RELATIVE CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Action of the relative class on cycles over closed subschemes (relative-cycles chapter, Lemma lemma-action-closed, canonical tag 0H6A). cited at proof lines 2461-2462. Gives [X/X/Y]_e cap beta = [W x_Y X]_{r + e} for beta = [W].
- Flat pullback of cycles (Chow Homology, Definition chow-definition-flat-pullback, canonical tag 02RB). cited at proof lines 2463-2464. Gives f^*beta = [f^{-1}(W)]_{r + e}, the same cycle.

Source reference. Original lemma 0H6A: sources/stacks-chow/tags.txt, source lines 21415–21415. Cited at proof lines 2461-2462: computes [X/X/Y]_e cap beta = [W x_Y X]_{r+e}.

Source reference. Original definition 02RB: sources/stacks-chow/tags.txt, source lines 3618–3618. Cited at proof lines 2463-2464: computes f^*beta = [f^{-1}(W)]_{r+e}.

Lemma 0.2. *Let (S, δ) and $f : X \to Y$ be as above. Assume f is flat of relative dimension e. For $\beta \in Z_r(Y)$ we have $f^*\beta = [X/X/Y]_e \cap \beta$ in $Z_{e+r}(X)$.*

Proof. It suffices to prove the equality locally on X. Thus we may assume Y is affine. In this case β is a finite integer linear combination of prime cycles. Since both sides of the equality are additive in β we may assume $\beta = [W]$ where W is an integral closed subscheme of Y of δ-dimension r. Let $f^{-1}(W) = W \times_Y X$ be the scheme theoretic inverse image of W in X. By Lemma 0H6A we have $[X/X/Y]_e \cap \beta = [W \times_Y X]_{r+e}$ and by Chow Homology, Definition 02RB we have $f^*\beta = [f^{-1}(W)]_{r+e}$. Thus we have the desired equality. □

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.